MILITARY SPECIFICATION SHEET
GUIDANCE CONTROL SECTION, GUIDED MISSILE
SERVO SECTION

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the servo section described herein shall consist of this specification and the latest issue of MIL-G-85742.

1. SCOPE

1.1 Scope. This specification sheet establishes the performance and test requirements for the servo section used in the Guidance Control Section of the SIDEARM Missiles.

2. APPLICABLE DOCUMENTS.

2.1 Government drawings. The following standard drawings, in addition to the applicable documents listed in MIL-G-85742, are applicable to this specification sheet.

Naval Air Systems Command
(Code Merit 30003)

1466As1202	Servo Section A3.
1466As1220	Power Converter Assembly A3A1.
1466AS1302	Power Supply Assembly.

FSC 1420

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3. REQUIREMENTS

3.1 Item description. The servo section (Drawing 1466AS1202) consists of the power converter (Drawing 1466AS1220), the gas grain assembly, the alternator, and the fin torque pistons. The power converter supplies power to the receiver section and provides launch lock/contact destruct functions. After power changeover (PCO), the launch lock function is used as part of the arming sequence for the warhead. The contact destruct funtion is used for detonating the warhead upon impact.

3.2 Characteristics.

3.2.1 Performance. Performance characteristics of the servo section shall be as specified herein.

3.2.1.1 Telemetry and aircraft signal line resistance. The resistance of the telemetry and aircraft signal lines shall conform to Table I.

3.2.1.2 Gas generator squib resistance. The squib circuit resistance shall be 3.25 tO.50 ohms.

3.2.1.3 Input voltage (pre-PCO). The power converter shall be able to operate on voltages meeting the following requirements.

3.2.1.3.1 Alternating current (ac) voltage. The input ac voltage shall meet the requirements specified in MIL-STD-704, DOD-STD-1399/103 with an applied voltage of 112 ± 8 volts root-mean-square (Vrms) at 400 ± 20 hertz Hz).

3.2.1.3.2 . Direct current (dc) voltage. The input dc voltage shall meet The requirements specified in MIL-STD-704 with an applied voltage of 28 volts (v).

3.2.1.4 Input voltage from alternator (post-PCO). The input voltage supplied by the alternator shall meet the following requirements when an external air pressure of 1150 ± 50 pounds per square-inch (psi) is applied.

3.2.1.4.1 Alternator voltage. After PCO anti removal of the aircraft 400 Hz, the alternator voltage shall be greater than 145 Vrms under full load.

3.2.1.4.2 Alternator frequency. Under the same conditions described in 3.2.1.4.1, the alternator frequency shall be 7,700 - 500, +700 Hz.

3.2.1.5 Output ac voltage. The output ac voltage of the power converter shall be 138 ± 7 Vrms at a frequency of 8.0 ± 0.3 kilohertz over the full input voltage range (see 3.2.1.3).

3.2.1.6 Cage voltage. The cage voltage shall be 9 ± 1 V when driving a 3 kilohm (kohm) load.

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3.2.1.7 G-Bias/Nav Gain (GB/NG) input disable voltage. Prior to PCO, the GB/NG input disable voltage shall be greater than 12 V.

3.2.1.8 Unregulated +6 V. The +6 V unregulated telemetry voltage shall be 6 + 1 V.

3.2.1.9 Regulator output. The regulator output current shall actuate the relay of Figure 1 at an alternator frequency of 6100 ± 400 Hz.

3.2.1.10 Launch lock voltage. The launch lock voltage shall be greater than or equal to 12 volts direct current when driving a 29.5 ± 2.5 ohm load. The ac ripple shall be less than 2 volts peak-to-peak.

3.2.1.11 Contact destructs No less than 4 seconds after PCO, the energy within the contact destruct fire pulse shall be not less than 33,000 ergs.

3.2.2 Physical characteristics. Physical characteristics shall be in accordance with Drawings 1466AS12U2 and 1466AS1220.

4. QUALITY ASSURANCE PROVISION

4.1 General. Each servo section shall meet the requirements of 3.2 when inspected and tested as specified on Drawings 1466ASJ202 and 1466AS1220, and the following paragraphs. Failure to meet any of the requirements shall be cause for rejection of the servo section.

4.1.1 Test environment. Unless otherwise specified, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): 23 ± 5 degrees Celsius (°C).

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.1.2 Test equipment accuracy. Unless otherwise specified test equipment accuracy shall be as specified in MIL-G-85742.

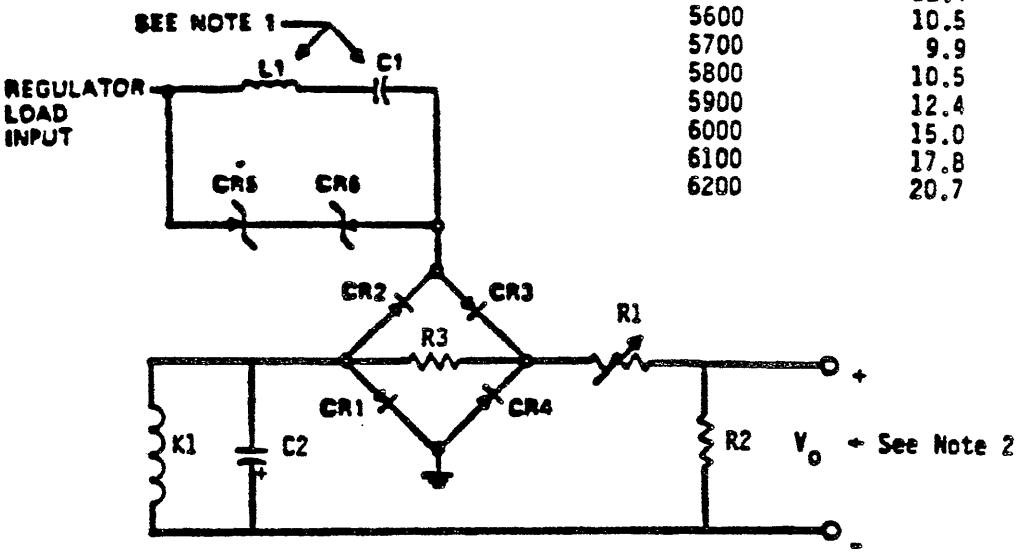
4.2 Quality conformance inspections

4.2.1 Unit test. The assembly shall be subjected to a unit test in order to establish conformance to the requirements specified herein.

4.2.2 Test methods. Details of methods and procedures shall be in accordance with the Government approved test procedure. The following test shall be performed with a simulated power supply assembly (Drawing 1466AS1302) connected, closing the feedback loop.

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FREQUENCY Hz	VOLTAGE RMS
5200	21.5
5300	18.1
5400	15.0
5500	12.4
5600	10.5
5700	9.9
5800	10.5
5900	12.4
6000	15.0
6100	17.8
6200	20.7



CIRCUIT COMPONENTS:

R1: 50 OHMS, ADJUSTABLE, 2 WATTS (W) (SEE NOTE 2)
R2: 1 OHM, 1 PERCENT, 1/4W
R3: 1.2 kOHMS, 5 PERCENT, 2W
K1: RELAY; ALLIED MRX-4213 OR EQUIV. (59A44A320)
250 ± 25 OHMS COIL RESISTANCE
C1: APPROXIMATELY 0.009 MICROFARADS (µF); 300 Vdc; PER MIL-C-5.
C2: 22 µF; 100V
CR1,CR2,CR3,CR4: IN4004 OR EQUIV.; FORWARD CURRENT AT 1V, 400 mA; INVERSE CURRENT AT 300V, 0.2 MICROAMPERES (µA); PIV-300V
CR5, CR6,: ZENER DIODE, IN3014, 180V, 10W
LI: TORODIAL CORE, A206068-2; 1160 TURNS NOMINAL, NO. 32 WIRE; RESISTANCE, NOMINAL 16 OHMS; INDUCTANCE, 78 TO 82 MILLIHENRIES (mH) (REMOVE TURNS TO OBTAIN INDUCTANCE VALUE.)

NOTES:

1. THE CIRCUIT SHALL BE TUNED TO 5700 ± 50 Hz WITH 15 Vrms OUTPUT BY SELECTION OF C1.

2. FOR AN OUTPUT VOLTAGE OF 25 MILLIVOLTS DIRECT CURRENT (mVdc) THE INPUT VOLTAGE SHALL BE WITHIN +2V OF THE VALUES GIVEN IN THE TABLE ABOVE, USING A SINE WAVE VOLTAGE GENERATOR. RESISTOR R1 MAY BE ADJUSTED AT THE FREQUENCY OF 5700 ± 15 Hz TO OBTAIN THE VALUES GIVEN IN THE TABLE.

FIGURE 1. Simulated interlock relay circuit.

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TABLE I. Telemetry and aircraft signal line resistance.

Signal name	From Connector - pin #	To Connector - pin #	Resistance
Polar Head	A3A1W1J1 - 2	A3A1J1 - 30	Less than 1 ohm
AZ Slaving	- 3	- 24	Less than 1 ohm
Az Slaving	- 3	A3A1W1P1 - 14	Less than 1 ohm
El Slaving	- 4	- 13	Less than 1 ohm
Roll Angle	- 5	- 1	Less than 1 ohm
AFC Voltage	- 6	- 12	Less than 1 ohm
Cage Enable	- 10	A3A1J1 - 23	Less than 1 ohm
El Solenoid	- 11	A3A1W1P1 - 6	16.8 ±0.4 kohms
El Solenoid	- 12	- 6	16.8 ±0.4 kohms
Az Solenoid	- 13	- 7	16.8 ±0.4 kohms
Az Solenoid	- 14	- 7	16.8 ±0.4 kohms
+175 Magamp	- 15	A3A1J1 - 25	Less than 1 ohm
Disable			Less than 1 ohm
El Torque	- 16	A3A1W1P1 - 10	Less than 1 ohm
Az Torque	- 17	- 11	Less than 1 ohm
Power Changeover	- 23	A3A1J1 - 7	Less than 1 ohm
Power Changeover	- 23	A3A1W1P2 - 9	Less than 1 ohm
Magamp In Az	A3A1W1J2 - 2	- 5	Less than 1 ohm
El G-Bias	- 5	A3A1W1P1 - 4	Less than 1 ohm
PRF/2	- 6	- 8	Less than 1 ohm
Magamp In El	- 8	A3A1W1P2 - 3	Less than 1 ohm
El Ref.	- 9	A3A1J1 - 18	10.0 ±0.5 kohms
El Ref.	- 9	A3A1W1P1 - 3	10.0 ±0.5 kohms
Atten. Control	- 10	- 16	Less than 1 ohm
Composite AGC	- 15	- 2	Less than 1 ohm
Az G-Bias	- 16	- 5	Less than 1 ohm
Pilot Headtone	- 19	A3A1J1 - 33	Less than 1 ohm
Error Signal	- 22	A3A1W1P1 - 9	Less than 1 ohm
Angle Boxcar	- 23	- 15	Less than 1 ohm
Launch Latch	A3A1J1 - 32	A3A1W1P2 - 6	Less than 1 ohm
Missile ID	- 17	A3A1J1 - 26	Less than 1 ohm

4.2.2.1 Gas generator squib resistance. Measure squib resistance between pins 22 and 31 on connector A3A1J1 to determine conformance to 3.2.1.2 using a test current of less than 5 milliamperes.

4.2.2.2 Input voltages (pre-PCO). The input ac voltage applied at pin 34 with return at pin 11 of A3A1J1, shall meet the requirements of 3.2.1.3.1. The input dc voltage applied to pin 17 of A3A1J1 shall meet the requirments of 3.2.1.3.2.

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4.2.2.3 Input voltage from alternator (post-PCO).

4.2.2.3.1 Servo air. Gas pressure shall be 1150 + 50 psi with a maximum dew point of not greater than -35°C. The particle size in the servo air shall be not more than 30 microns. The flow rate shall be approximately 30 cubic feet per minute at standard temperature and pressure.

4.2.2.3.2 Alternator voltage. Apply servo air in accordance with 4.2.2.3.1 and remove aircraft 400 Hz. Measure the alternator voltage at pin 21 of connector A3A1W1J1 using pin 20 as ground. The voltage shall meet the requirements of 3.2.1.4.1.

4.2.2.3.3 Alternator frequency. The frequency of the ac voltage in 4.2.2.3.2 shall meet the requirements of 3.2.1.4.2.

4.2.2.4 Output ac voltage. Apply an input voltage in accordance with 3.2.1.3. Measure the voltage and frequency between pins 24 and 9, pins 25 and 9, and pins 8 and 5 of connector A3A1W1J1. The output voltages shall meet the requirements of 3.2.1.5.

4.2.2.5 Cage voltage. Connect a 3 kohm resistor between pins 1 and 31 of connector A3A1J1. Measure the cage voltage across the 3 kohm load. The voltage shall meet the requirements of 3.2.1.6.

4.2.2.6 GB/NG input disable voltage. With an input voltage in accordance with 3.2.1.3.2, measure the GB/NG input disable voltage on pin 14 of A3A1W1J2 with return on pin 4. The voltage shall meet the requirements of 3.2.1.7.

4.2.2.7 Unregulated +6 V. Apply an input voltage in accordance with 4.2.2.2. Measure the telemetry voltage on pin 4 of A3A1W1P2 with return on pin 1. The voltage shall meet the requirements of 3.2.1.8.

4.2.2.8 Regulator output. Apply input voltages in accordance with 4.2.2.2. Connect the circuit of Figure 1 to pin 2 of A3A1J1 using pin 31 as ground. Apply servo air in accordance to 4.2.2.2.1. Measure the alternator frequency between pins 20 and 21 of A3A1W1J1. The output shall meet the requirements of 3.2.1.9.

4.2.2.9 Launch lock voltage. Connect a 29.5 ± 2.5 ohm resistor between pins 7 and 1 on A3A1W1P2. With servo air applied in accordance with 4.2.2.3.1, measure the launch lock voltage across the 29.5 ± 2.5 ohm load. The voltage shall meet the requirements of 3.2.1.10.

4.2.2.10 Contact destruct. With servo air applied in accordance with 4.2.2.3.1, measure the energy within the fire pulse between pins 2 and 1 of A3A1W1P2 when E8 is shorted to E9. The output shall meet the requirements of 3.2.1.11.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85742/5(AS)	Servo Section

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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